

Pump P-101 Equipment Manual

1. Document Information

Equipment Tag: P-101
Equipment Type: Centrifugal Process Pump
Service: Industrial process-fluid transfer
Document Type: Equipment Technical Manual
Document Status: Synthetic Demonstration Document
Revision: 1.0

Note: This document contains synthetic equipment information created for the Sovereign AI Workbench demonstration. It does not represent a real industrial asset.

2. Equipment Description

Pump P-101 is a centrifugal process pump used to transfer industrial process fluid between process units. The pump is driven by an electric motor and is designed for continuous operation under the specified operating conditions.

The major components of P-101 include:

- Electric motor
- Pump casing
- Impeller
- Shaft
- Bearings
- Mechanical seal
- Coupling
- Suction connection
- Discharge connection

3. Normal Operating Specifications

Parameter	Normal Operating Value
Rated Flow	120 m³/h
Rated Head	45 m
Motor Power	30 kW
Operating Speed	1450 RPM
Normal Temperature	60–75 °C

Parameter	Normal Operating Value
-----------	------------------------

Normal Discharge Pressure	4.0–4.5 bar
---------------------------	-------------

Normal Vibration	Below 4.5 mm/s RMS
------------------	--------------------

The pump should normally operate within the specified ranges. Continuous operation outside these limits requires investigation.

4. Vibration Monitoring Limits

Vibration is a key condition-monitoring parameter for P-101.

Vibration Level	Condition	Required Action
-----------------	-----------	-----------------

Below 4.5 mm/s RMS	Normal	Continue monitoring
--------------------	--------	---------------------

4.5–7.1 mm/s RMS	Warning	Inspect and investigate
------------------	---------	-------------------------

Above 7.1 mm/s RMS	Critical	Immediate investigation and controlled response
--------------------	----------	---

A sustained increase in vibration may indicate mechanical problems such as bearing degradation, coupling misalignment, imbalance, looseness, or other rotating-equipment abnormalities.

A single abnormal sensor reading should be verified before taking major maintenance action.

5. Temperature Monitoring

The normal operating temperature range for P-101 is **60–75 °C**.

A temperature increase should be investigated when it occurs together with abnormal vibration or other changes in operating conditions.

Possible causes of abnormal temperature include:

- Bearing problems
 - Insufficient lubrication
 - Excessive mechanical friction
 - Process or operating-condition changes
 - Cooling problems
-

6. Pressure Monitoring

The normal discharge pressure range is **4.0–4.5 bar**.

Abnormal pressure may indicate:

- Restricted flow

- Valve problems
- Suction conditions outside the normal range
- Process changes
- Pump performance degradation

Pressure readings should be evaluated together with flow, speed, temperature, and vibration.

7. Common Causes of High Vibration

High vibration in P-101 may be associated with:

1. Bearing degradation or damage
2. Coupling misalignment
3. Shaft imbalance
4. Mechanical looseness
5. Impeller imbalance or damage
6. Foundation or mounting problems
7. Operating conditions outside the design range

High vibration should not automatically be attributed to one cause. Inspection and supporting evidence are required before identifying the root cause.

8. Condition Monitoring Procedure

The following parameters should be monitored during normal operation:

- Vibration
- Bearing temperature
- Pump discharge pressure
- Pump flow
- Motor speed
- Motor operating condition

Sensor readings should be recorded at regular intervals and compared with the normal operating limits.

A sudden change or sustained deviation should be reported for further investigation.

9. Response to High Vibration

If P-101 vibration enters the warning range:

1. Verify the sensor reading.
2. Check whether the abnormal value persists.
3. Review recent operating conditions.
4. Check temperature and pressure readings.
5. Review recent maintenance history.
6. Inspect the pump, coupling, and bearing condition according to the applicable SOP.
7. Escalate the condition if vibration continues to increase.

If vibration exceeds the critical limit, the responsible operator should follow the applicable safety and controlled-shutdown procedure.

10. Maintenance Considerations

Regular inspection should include:

- Bearing condition
- Lubrication condition
- Coupling alignment
- Shaft condition
- Mechanical seal condition
- Pump mounting and foundation
- Impeller condition

Maintenance activities and abnormal observations should be documented in the equipment maintenance record.

11. Safety Precautions

Before performing physical inspection or maintenance:

- Follow site safety procedures.
 - Isolate the equipment when required.
 - Follow lockout/tagout procedures where applicable.
 - Confirm that rotating components have stopped before physical inspection.
 - Use appropriate personal protective equipment.
 - Do not bypass equipment protection or safety interlocks.
 - Escalate unsafe conditions to the responsible personnel.
-

12. Example Condition-Monitoring Event

The following example is provided for demonstration purposes.

A sensor monitoring event records a vibration value of **8.7 mm/s RMS** for P-101.

According to the limits in this manual:

8.7 mm/s RMS > 7.1 mm/s RMS

Therefore, the reading is classified as **Critical** and requires investigation.

The sensor reading alone does not establish the root cause. The maintenance history, SOP, incident records, and physical inspection results should be reviewed before determining the cause.

13. Relationship to Sensor Analytics

The Sovereign AI Workbench may receive sensor data for P-101 through CSV files.

For example:

Vibration: 8.7 mm/s RMS

Temperature: 71 °C

Pressure: 4.1 bar

Speed: 1450 RPM

The Data Analytics module can identify abnormal sensor behavior and calculate an equipment health score.

The equipment manual provides the technical limits used to interpret these measurements.

14. Document Control

Equipment: P-101

Document: Equipment Technical Manual

Revision: 1.0

Purpose: Synthetic demonstration and AI/RAG testing

Confidentiality: Demonstration data only

End of Document